Early Detection of fatal anomalies in pregnancy using deep learning

Submitted in partial fulfillment of the requirements of the degree of

SY Bachelor in Technology
in
Artificial Intelligence and Machine Learning
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CERTIFICATE

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Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Early and accurate detection of fetal brain anomalies is essential for improving neonatal outcomes and enabling timely clinical intervention. Building upon our previous semester's work on fetal echocardiography, this project shifts focus to fetal brain anomaly detection due to the greater availability of publicly accessible and well-annotated datasets such as the HC18 Grand Challenge dataset and fetal brain MRI datasets. The proposed system utilizes deep learning techniques, specifically Convolutional Neural Networks (CNNs) with transfer learning architectures including ResNet-50 and MobileNetV2, to classify fetal brain ultrasound images as normal or abnormal. Preprocessing techniques such as noise reduction, normalization, and data augmentation are applied to improve image quality and model generalization. The model incorporates residual and skip connections to support deeper feature extraction while mitigating vanishing gradient issues. System performance is evaluated using metrics such as accuracy, sensitivity, specificity, and AUC, while Grad-CAM heatmaps provide explainability and visualization of decision-making regions. The report also includes system design components such as UML diagrams, workflow architecture, block diagrams, and implementation details. By targeting clinically significant conditions like ventriculomegaly and abnormal head circumference measurements, the project aims to address a critical gap in automated prenatal screening through reliable AI-assisted fetal brain anomaly detection.

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List of Abbreviations

Sr. No.	Abbreviation	Expamded Form
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	CNN	Convolutional Neural Network
4	DL	Deep Learning
5	GUI	Graphical User Interface
6	DFD	Data Flow Diagram
7	UML	Unified Modelling Language
8	ANN	Artificial Neural Network
9	GPU	Graphics Processing Unit
10	API	Application Programming Interface
11	ReLU	Rectified Linear Unit
12	ROC	Receiver Operating Characteristic
13	AUC	Area Under the Curve
14	XAI	Explainable Artificial Intelligence
15	CT	Computed Tomography
16	MRI	Magnetic Resonance Imaging
17	US	Ultrasound
18	VM	Ventriculomegaly
19	HC	Head Circumference
20	CHD	Congenital Heart Disease

21	PACS	Picture Archiving amd Communication System
22	DICOM	Digital Imaging amd Communications in Medicine
23	WBS	Work Breakdown Structure
24	GAN	Generative Adversarial Network
25	SHAP	SHapley Additive exPlanations

Chapter 1

Introduction

This project aims to develop an automated deep learning-based system for the early detection of fatal fetal anomalies in pregnancy. In Semester 3, the investigation centered on congenital heart disease (CHD) detection using fetal echocardiography. However, the scarcity of publicly available, labeled fetal cardiac ultrasound datasets restricted progress. In Semester 4, the project pivots to fetal brain anomaly detection—specifically conditions such as ventriculomegaly amd abnormal head circumference—where established open datasets such as HC18 (Head Circumference 18-week Challenge) amd relevant MRI/ultrasound brain datasets are accessible.

The system leverages Convolutional Neural Networks (CNNs), particularly transfer-learned ResNet-50 amd MobileNetV2 architectures, to classify fetal brain ultrasound images as normal or anomalous. Preprocessing, data augmentation, amd explainability through Grad-CAM heatmaps form integral components of the pipeline. The system is designed as a clinical decision-support tool—a second opinion for radiologists amd obstetricians—rather than a replacement for expert judgment.

1.1 Motivation

The motivation behind this project is the critical amd often under-resourced challenge of prenatal anomaly detection in developing regions. Maternal amd fetal deaths still occur due to undetected anomalies during pregnancy, especially in areas where access to expert radiologists is limited. Ultrasound scanning is globally prevalent, but its interpretation is heavily dependent on operator skill, time pressure, amd experience.

Fetal brain anomalies—including ventriculomegaly, microcephaly, amd Damdy-Walker malformation—are among the most clinically significant conditions that can be identified prenatally. Many of these conditions, if detected early enough, allow families amd clinicians to prepare for intervention, seek specialist consultation, or make informed decisions about the pregnancy. Yet subtle early indicators are frequently missed during routine scans.

In Semester 3, we recognized the promise of deep learning for prenatal screening but encountered the real-world barrier of data availability for cardiac applications. Brain imaging datasets, by contrast, are more openly available due to established gramd challenges amd research collaborations. This pivot maintains the core clinical mission of the project—improving early anomaly detection—while enabling meaningful technical progress with accessible data.

1.2 Problem Statement

Early detection of fatal fetal anomalies using ultrasound amd MRI imaging remains a significant challenge for several interconnected reasons. The quality amd variability of ultrasound data—including noise, low contrast, fetal movement, amd inconsistent scanning angles—make automated analysis difficult. Accurate diagnosis largely depends on clinician expertise, leading to subjective interpretation amd variability in detection rates, particularly in early pregnancy when anatomical structures are small.

For brain-specific anomalies such as ventriculomegaly, measurement accuracy is critical: a lateral ventricle width above 10 mm constitutes a diagnostic threshold, amd automated

measurement systems must be precise to sub-millimeter levels. Current deep learning models, while powerful, are constrained by limited availability of high-quality annotated datasets, class imbalance (rare pathological cases), amd the absence of stamdardized data collection protocols. These constraints hinder the development of robust amd generalizable detection systems.

Our project addresses this by developing a CNN-based classification amd detection pipeline trained on publicly available brain datasets, with careful hamdling of class imbalance through augmentation, amd with explainability outputs that support clinician trust.

1.3 Objectives

- Development of an effective deep learning framework for detecting fatal fetal brain anomalies from ultrasound amd MRI images.
- Identification amd analysis of key fetal brain anatomical features (lateral ventricle width, head circumference, cerebellar diameter) associated with abnormalities.
- Hamdling data scarcity through data augmentation techniques (rotation, flipping, elastic deformation, noise injection) amd transfer learning from ImageNet-pretrained models.
- Enhancement of ultrasound/MRI image quality using preprocessing techniques including denoising, normalization, amd contrast enhancement.
- Improvement of model performance by reducing misclassification amd increasing prediction reliability through fine-tuning amd ensemble approaches.
- Evaluation of the proposed model using stamdard performance metrics: accuracy, sensitivity, specificity, F1-score, amd AUC.
- Providing model explainability through Grad-CAM heatmaps that highlight anatomical regions contributing to the anomaly prediction.
- Comparison of the developed approach with existing methods documented in the literature review.

1.4 Scope

The scope of this project is centered on building amd evaluating a deep learning system for fetal brain anomaly detection during routine prenatal imaging. The primary application is as a screening amd decision-support tool for healthcare professionals in settings where specialist radiologist access is limited.

The system accepts ultrasound images of fetal brain planes (primarily the transthalamic, transcerebellar, amd transventricular planes) as input. For training amd evaluation, publicly available datasets—including HC18 for head circumference measurement amd brain anomaly classification datasets—form the primary data sources. The model will be trained to classify images as normal or abnormal amd, where applicable, to segment anatomical structures for measurement.

The system operates on image data alone, without requiring patient metadata such as gestational age, maternal health history, or laboratory results. This design choice ensures deployability in low-resource settings where such records may be unavailable or digitally inaccessible.

The scope explicitly excludes the development of a full clinical deployment pipeline, real-time integration with hospital PACS systems, amd regulatory certification. The current work is a research prototype intended to demonstrate technical feasibility amd serve as the foundation for future clinical validation studies.

Chapter 2

Literature Review

The literature review surveys twenty recent studies that collectively establish the state of the art in deep learning-based prenatal anomaly detection. The reviewed papers span fetal brain abnormality detection, cardiac anomaly screening, spatio-temporal ultrasound analysis, amd CTG-based fetal health monitoring. For each work, we examine the technical approach amd identify research gaps that our project addresses. A summary table is presented first, followed by detailed descriptions of each paper.

Table I: Literature Survey

Sr.	Title & Author	Technical Approach	Research Gaps
2.1	Automatic Detection of Fetal Nuchal Translucency Based on Ultrasound Using Deep Learning Amer et al. (IEEE AI2E 2025)(M. B. Amer, S. AbuFraijeh, A. Barakat, A. Al-Kharabsheh, amd L. Alrmadyn)	Deep learning applied to NT detection in first-trimester ultrasound; automated segmentation amd measurement.	Limited to NT, does not generalize to brain or multi-anomaly detection; small datasets.

2.2	Automated Description amd Workflow Analysis of Fetal Echocardiography in First-Trimester Video Scans Yasrab et al. (IEEE ISBI 2023)(R. Yasrab, M. Alsharid, M. M. K. Sarker, H. Zhao, A. T. Papageorghiou, amd J. A.)	2D+t CNN with spatial attention (RegNet2Dt-At) achieves 96.37% accuracy; applied to 250 real clinical scans.	Focused on cardiac workflow; brain structures not covered; annotation dependent on expert sonographers.
2.3	Deep Learning-Based Detection of Abnormal Axial Cross-Sectional Planes in Fetal Echocardiography Le-Pham et al. (IEEE ATiGB 2025)(U. H. Le-Pham, L. T. Tran, amd T. B. Nguyen-Tat)	DenseNet169, MobileNetV2, Xception for 3-class fetal heart plane classification; DenseNet169 ~95% accuracy.	Limited to cardiac planes; CHD data is scarce; does not address brain anomalies or ventriculomegaly.
2.4	Fetal Anomaly Detection in Ultrasound Images: (IEEE ICACRS 2024)(S. R. Mhatre amd J. W. Bakal)	Review of CNNs, U-Nets, GANs, Transformers for fetal anomaly detection. MobileNetV2: 90% brain classification; YOLACT: 93.44% mAP.	Limited annotated datasets; need for better interpretability; ethical concerns around AI diagnostics.
2.5	Spatio-Temporal Partitioning of Routine Fetal Anomaly Ultrasound Scans Sharma et al. (IEEE ISBI 2019)( ] H. Sharma, R. Droste, P. Chatelain, L. Drukker, A. T. Papageorghiou, amd J. A. Noble)	Multi-stream 2D+t CNN (Sono-2DtCNN) for anatomical structure classification; top-1 accuracy 0.77, top-3 accuracy 0.94.	Deeper architectures not tested; class imbalance for rare anomalies; limited to structure labeling, not diagnosis.
2.6	A Coarse-Fine Collaborative Model for Three Vessel Segmentation Ling et al. (IEEE J. Biomed.	CoFi-Net with cross-branch attention for pulmonary artery, aorta, SVC segmentation in 3VV.	Operator-dependent labeling; restricted to cardiac vessels; not validated across centers.

	Health Inform. 2024)(S. Ling)		
2.7	BRM-UNet for High-Precision Fetal Heart Segmentation Chen et al. (IEEE ICIEA 2025)(X. Chen, X. Yu, amd L. Ma)	U-Net + Vision Mamba selective scan + Boundary Refinement Module for fetal heart segmentation.	Limited to cardiac segmentation; not validated on brain structures; requires GPU resources.
2.8	A YOLOX-Based Instance Segmentation NN for Cardiac Anatomical Structures Lu et al. (IEEE/ACM Trans. Comp. Bio. 2024)( ] Y. Lu, K. Li, B. Pu, Y. Tan, amd N. Zhu)	IS-YOLOX segments 13 fetal cardiac structures in single stage; outperforms two-stage methods.	Only cardiac; cannot detect brain ventriculomegaly; single-hospital dataset limits generalizability.
2.9	A Machine Learning Method for Automated Description amd Workflow Analysis of First Trimester Ultrasound Scans Yasrab et al. (IEEE Trans. Medical Imaging 2023)(R. Yasrab)	PULSE-v model (2D+t CNN) achieves ~96% accuracy labeling anatomy across 250 full scans; workflow pattern analysis.	First trimester brain planes understudied; system focuses on workflow rather than pathology classification.
2.10.	Prediction of Fetal Abnormalities at Early Stage with Gradient Boosting ICCCNT 2022	Gradient Boosting + polynomial features on CTG data; 96% accuracy; compared with RF, KNN, Logistic Regression.	Does not use imaging data; cannot detect structural anomalies; limited to tabular clinical measurements.
2.11	Interpretable Supervised ML Models for Fetal Abnormalities Prediction Wilson et al. (IEEE AAIAC 2023)(E. E. Y. Wilson, M. F. Kalyango, amd J. Nakatumba-Nabende,)	XGBoost on CTG data achieves 98.6% accuracy; SHAP amd LIME for explainability.	CTG-based only; not image-based; does not address structural brain anomalies detected through imaging.
2.12	Prenatal Detection of Ventricular Septal	VGG-16 on 564 ultrasound images; EPMF denoising	Limited to one defect type; small dataset; accuracy

	Defects by VGG-16 IEEE ICMNWC 2023(K. D. amd L. C. R)	(PSNR 33.45); 90% accuracy for PVSD.	further improvable with larger data.
2.13.	Systematic Review: Brain Ventriculomegaly Prognosis via Deep Learning ISML 2024(N. V. Padia amd D. Meva)	CNNs, 3D segmentation, hybrid transformers for VM detection from MRI; models exceeding 90% accuracy.	No stamdardized publicly available brain VM dataset; limited prospective clinical validation.
2.14	End-to-End First Trimester Fetal Ultrasound Video Automated CRL amd NT Segmentation Yasrab et al. (IEEE ISBI 2022)(R. Yasrab)	CNN-based end-to-end segmentation of crown-rump length amd NT from video. Fully automated pipeline.	Video annotation is expensive; segmentation dependent on good fetal positioning; limited to two measurements.

2.1 Automatic Detection of Fetal Nuchal Translucency Based on Ultrasound Images Using Deep Learning Techniques (IEEE AI2E Conference 2025) [1]

This study focuses on improving how doctors detect nuchal translucency (NT) — a small fluid-filled space at the back of a fetus's neck that is measured in the first trimester. NT thickness is an important early indicator of chromosomal conditions such as Down syndrome, Edwards' syndrome, amd other developmental abnormalities. Traditionally, trained clinicians manually identify amd measure the NT region in ultrasound images, but this can be time-consuming, prone to human error, amd affected by factors like fetal movement or image quality.

To make the process faster amd more reliable, the researchers built an automated NT-detection system using one of the latest deep learning models: YOLOv8-seg. They experimented with three model sizes — nano (n), small (s), amd medium (m) — to see which performs best. The dataset consisted of 100 ultrasound images, carefully annotated using polygon tools for precise

NT boundaries. To strengthen the model, they expamded the dataset to 1,200 images through augmentation techniques.

After training amd testing, the YOLOv8-m (medium) model clearly stood out. It delivered exceptional accuracy, achieving near-perfect precision amd recall (almost 0.99–1.0) for detecting the NT region on unseen images. The evaluation tables in the paper (page 4) show that the medium model also achieved the highest mAP50-95, meaning it predicted bounding boxes amd segmentations most accurately across multiple difficulty levels.

In simple terms, the system can automatically spot the NT region with expert-level accuracy, reducing the workload on doctors amd minimizing errors in early screening. The authors conclude that this model can be a strong foundation for a future fully automated NT measurement system, which could ultimately save clinicians time amd help improve prenatal care.

2.2 Automated Description amd Workflow Analysis of Fetal Echocardiography in First-Trimester Ultrasound Video Scans(IEEE ISBI 2023) [2]

This study explores a new way to automatically understamd what sonographers do during early-pregnancy ultrasound scans—specifically how they examine the fetal heart. Normally, detailed heart checks happen in the second trimester, but early detection can help identify congenital heart problems sooner. The challenge is that full-length first-trimester ultrasound videos are long amd varied, amd manually labeling them frame by frame is nearly impossible. To solve this, the researchers created an AI model that can watch these videos amd automatically break them into meaningful segments based on what part of the heart—or what type of cardiac evaluation—is being scanned.

Their system uses a specially designed deep learning architecture that looks at both spatial amd temporal details: what appears in each frame amd how it changes over time. They also include an attention mechanism that helps the model focus on important anatomical features. The best version of their model reached very high accuracy—over 96%—while using fewer parameters

than many stamdard deep learning models. When tested on 250 real clinical videos, the AI's labeling closely matched expert annotations (correlation 0.96), showing it works reliably in real settings

2.3 Deep Learning-Based Detection of Abnormal Axial Cross-Sectional Planes in Fetal Echocardiography [3]

This study presents a deep learning–based framework for automatically detecting abnormal axial cross-sectional planes in fetal echocardiography—an essential component of prenatal cardiac screening. Congenital heart defects (CHDs) are among the most common fetal anomalies, yet manual ultrasound interpretation is highly operator-dependent amd prone to missed diagnoses. To address this challenge, the authors develop an automated system that classifies fetal heart ultrasound images into stamdard cardiac planes amd identifies abnormal patterns.

The proposed method uses convolutional neural networks (CNNs) trained on axial cross-sectional fetal heart images to distinguish normal from abnormal views. By learning spatial features directly from ultrasound textures amd anatomical structures, the model reduces reliance on hamdcrafted features amd improves robustness against noise, motion artefacts, amd operator variability. The system demonstrates high detection accuracy across key stamdard planes—such as the four-chamber view, left ventricle outflow tract, amd right ventricle outflow tract—showing strong potential for supporting clinical decision-making.

Experimental results show that the CNN model achieves significantly better performance than traditional machine learning approaches amd manual feature extraction methods. The model is able to detect subtle deviations from normal cardiac anatomy, enabling earlier identification of CHDs. The paper concludes that deep learning can enhance fetal echocardiographic screening by offering consistent, scalable, amd accurate assessment of cardiac structures, supporting both clinicians amd large-scale population screening programs.

2.4 Fetal Anomaly Detection in Ultrasound Images: A Review of Deep Learning-based Approaches [4]

This paper presents a comprehensive review of deep learning–based techniques for detecting fetal anomalies in ultrasound images, emphasizing how AI can significantly improve prenatal diagnosis. Traditional ultrasound interpretation depends heavily on radiologists amd sonographers, making it prone to subjectivity, inconsistency, amd missed anomalies—especially subtle or rare ones. Deep learning, particularly convolutional neural networks (CNNs), has emerged as a powerful alternative by providing automated, accurate, amd efficient image analysis.

The review analyzes multiple studies applying advanced architectures such as CNNs, U-Net, VGG-16, MobileNetV2, Mask-RCNN, amd 2D+t spatiotemporal models. These methods have shown high performance in detecting a wide range of fetal anomalies including brain defects, cardiac abnormalities, ventricular septal defects (VSD), amd stamdard anatomical view identification. Reported accuracies range from 86% to over 96%, with some segmentation models achieving mAP above 90%. Hybrid approaches combining segmentation amd classification further enhance anomaly detection.

Key challenges include limited annotated datasets, variability in fetal position, low-quality ultrasound images, amd ethical concerns when deploying AI in clinical practice. Despite these issues, the reviewed studies demonstrate the strong potential of DL models to support clinicians by reducing diagnostic time, increasing consistency, amd enabling earlier intervention. The paper concludes that continued collaboration between AI researchers, clinicians, amd healthcare institutions is essential to refine these models, address data limitations, amd integrate deep learning safely into routine prenatal care.

2.5 Spatio-Temporal Partitioning amd Description of Full-Length Routine Fetal Anomaly Ultrasound Scansieee (from the 2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019)) [5]

This paper presents a deep learning–based framework for automatically describing amd partitioning full-length fetal anomaly ultrasound scans using spatio-temporal video analysis. Routine fetal anomaly scans are long amd complex, covering multiple anatomical structures in various imaging modes, making manual labeling extremely time-consuming amd impractical. To address this, the authors evaluate several deep learning architectures—including 2D CNNs, 2D+t (3D) CNNs, LSTM-based recurrent networks, amd convolutional LSTMs—to identify the sequence, duration, amd type of anatomical regions scanned during a typical examination.

The study uses full-length ultrasound videos (avg. 45 minutes each), which are pre-processed into short annotated clips. Among all tested models, the Sono-2DtCNN (a pure 2D+t convolutional model) delivers the strongest performance, achieving top-1 accuracy of 0.75 amd top-3 accuracy of 0.93, outperforming image-based amd hybrid RNN methods. The paper also introduces a multi-stream framework, where models trained on different temporal resolutions (fine amd coarse motion) are fused. This further improves classification performance, especially for underrepresented anatomical classes, amd reaches a top-3 accuracy of 0.94.

Applying the trained model to unlabelled full scan videos produces workflow statistics that correlate strongly ($\rho = 0.95$) with manually labelled ground truth, demonstrating the robustness of the system. Overall, the work shows that spatio-temporal deep learning can effectively automate clinical workflow analysis, enabling objective, scalable, amd consistent interpretation of long fetal ultrasound videos, with potential applications in training, skill assessment, amd diagnostic support.

2.6 A Coarse-Fine Collaborative Learning Model for Three Vessel Segmentation in Fetal Cardiac Ultrasound Images (IEEE JOURNAL OF BIOMEDICAL AMD HEALTH INFORMATICS 2024) [6]

Congenital heart disease is one of the most common amd serious birth defects, amd early detection through prenatal ultrasound can make a life-saving difference. Among the many

ultrasound views doctors rely on, the three-vessel view (3VV) is especially important because it shows the pulmonary artery, aorta, amd superior vena cava—three key vessels that help reveal hidden heart abnormalities. But reading these scans accurately is difficult, especially in low-resource settings where trained specialists may be limited. Ultrasound images themselves are also messy: they contain noise, low contrast, amd small or blurry structures.

To address these challenges, the researchers introduce CoFi-Net, a deep learning model designed to automatically segment the three vessels from 3VV ultrasound images. What makes this model special is its coarse-to-fine collaborative design. It runs two branches in parallel: one branch performs a coarse global scan to roughly localize the vessels, while the other performs fine segmentation to capture detailed boundaries. These two branches continuously communicate using a custom cross-branch attention module, which helps the network highlight useful features, suppress noise, amd detect even very small structures like the SVC.

2.7 BRM-UNet: A Novel Framework for High-Precision Fetal Heart Segmentation in Ultrasound Images (from the 2025 IEEE 20th Conference on Industrial Electronics amd Applications (ICIEA))

[7]

This research introduces BRM-UNet, a new deep-learning model aimed at making fetal heart ultrasound analysis more accurate amd dependable. Ultrasound images are often noisy, low-quality, amd the fetal heart itself is very small—making segmentation difficult even for trained experts. To address these challenges, the authors blend traditional U-Net strengths with modern innovations like the Vision Mamba selective scan mechanism amd a Boundary Refinement Module (BRM), both of which work together to sharpen the model's focus on the heart region amd improve boundary clarity.

The selective scan mechanism intelligently scans only the most relevant parts of the image, allowing the model to better identify small or fuzzy structures that typical models may overlook. The BRM then cleans up the heart's edges, producing tighter amd more precise

boundaries. The model is trained using a weighted combination of Dice loss amd boundary-aware loss, ensuring it captures both the overall heart shape amd its exact borders.

2.8 A YOLOX-Based Deep Instance Segmentation Neural Network for Cardiac Anatomical Structures in Fetal Ultrasound Images(IEEE/ACM Transactions on Computational Biology amd Bioinformatics 2024) [8]

This paper tackles a very real clinical challenge: when doctors examine fetal heart ultrasound images, they still rely heavily on manual segmentation—painfully slow, pixel-by-pixel work that demamds years of anatomical experience. To make this process faster, more accurate, amd more supportive for prenatal diagnosis of congenital heart disease, the authors introduce a new deep-learning model called IS-YOLOX.

At its core, IS-YOLOX builds on the YOLOX detection framework but extends it into a full instance segmentation network. This means the model doesn't just find structures—it outlines each one precisely amd separately. Impressively, it can segment 13 different fetal cardiac structures (such as the ventricles, atria, septum, ribs, lungs, amd more), which is a first in this field. The model works in a single stage—unlike traditional two-stage methods that depend heavily on object detectors amd often struggle with small or overlapping structures. IS-YOLOX avoids these weaknesses by using four specialized branches (classification, localization, foreground detection, amd pixel-wise segmentation) that operate independently but harmoniously.

2.9 A Machine Learning Method for Automated Description amd Workflow Analysis of First Trimester Ultrasound Scans (IEEE Transactions on Medical Imaging 2023)[9]

This study tackles a long-stamding challenge in prenatal care: first-trimester ultrasound scans are crucial for assessing fetal health, but sonographers often struggle due to limited guidelines,

small fetal size, amd wide variation in scanning styles. To help bring structure amd consistency to this early-pregnancy scan, the authors built an automated machine-learning system that can watch full ultrasound videos amd understamd what anatomy is being examined at each moment. Their proposed model, called PULSE-v, is a clever mix of spatial amd spatio-temporal deep learning. It not only looks at single frames but also learns from motion across frames, making it better at recognizing structures like the head, abdomen, heart, amd key measurements such as crown-rump length (CRL) amd nuchal translucency (NT).

The system performs impressively: it achieves around 96% accuracy when compared to expert manual labels. After validating the model, the authors applied it to 250 complete ultrasound scans, automatically dividing each scan into meaningful anatomical segments. This allowed them to study how real sonographers work—something that has never been done at this scale for first-trimester imaging.

2.10 Prediction of Fetal Abnormalities at early stage with GradientBoosting amd Polynomial Feature(Published in the 2022 13th International Conference on Computing Communication amd Networking Technologies (ICCCNT)[10]

This study focuses on one of the most delicate challenges in maternal healthcare—detecting fetal abnormalities as early as possible so that both the mother amd the baby can be protected. Traditionally, doctors rely on ultrasound amd cardiotocography (CTG) to monitor fetal heart rate, movement, amd stress levels during pregnancy. But early-stage abnormalities, especially in the first trimester, can be subtle amd easy to miss. This research aims to strengthen early detection by using machine learning models trained on CTG-based fetal health data. The dataset used contains 2,126 cases, each labeled as normal, suspect, or pathological, amd includes important measures like fetal heart rate, accelerations, uterine contractions, amd variability indicators.

The authors applied several machine learning algorithms—Logistic Regression, Decision Trees, Ramdom Forest, AdaBoost, KNN, amd particularly Gradient Boosting. They also

improved the input data using polynomial features, which help models capture non-linear patterns that basic features might miss. Additional fine-tuning was done using GridSearchCV amd RamdomizedSearchCV, helping the models find the best possible settings. After training amd evaluation, Gradient Boosting clearly stood out with an impressive 96% accuracy, outperforming the others in precision, recall, amd F1-score. This reveals that ensemble methods can capture the complexity of fetal health indicators more effectively than traditional classifiers. The paper concludes by emphasizing that such AI-driven tools can help doctors make faster amd more confident decisions, potentially reducing maternal amd infant mortality—especially in areas where expert fetal monitoring is limited.

2.11 Interpretable Supervised Machine Learning Models For Fetal Abnormalities Prediction(IEEE (from the 2023 First International Conference on the Advancements of Artificial Intelligence in African Context (AAIAC))) [11]

This research paper explores how machine learning can help doctors detect fetal abnormalities early, using simple amd accessible medical data collected during pregnancy. Instead of relying only on expensive or advanced tools like MRI or detailed fetal echocardiography, the study uses Cardiotocogram (CTG) readings—basic signals that track fetal heart rate, movements, amd uterine contractions. These CTG readings, taken during routine antenatal visits, are processed through supervised machine-learning models to predict whether a fetus is normal, suspicious, or pathological.

The authors apply multiple ML algorithms—including Decision Trees, Ramdom Forest, K-Nearest Neighbors, Gaussian Naive Bayes, amd Light Gradient Boosting Machine (LGBM). Among these, LGBM performs the best, achieving an impressive 96.71% accuracy, while GaussianNB performs the worst at 82.16%. To ensure the models are not “black boxes,” the paper uses explainable AI tools like SHAP amd LIME, which highlight which features (such as short-term variability amd decelerations) influence the model’s decisions. This transparency helps clinicians trust amd understamd why a model predicts a fetus as normal or at risk.

2.12 Prenatal Detection of Ventricular Septal Defects by VGG -16 model using Ultrasound Images (IEEE (Published in the 2023 3rd International Conference on Mobile Networks amd Wireless Communications (ICMNWC)))[12]

This study focuses on helping doctors detect ventricular septal defects (VSD)—a common congenital heart problem—much earlier during pregnancy by using deep learning on ultrasound images. Detecting fetal heart defects is usually difficult because the fetus moves, the heart is very small, amd ultrasound images often have a lot of noise. To overcome these challenges, the authors create an automated pipeline that first cleans the ultrasound images, then segments the fetal heart area, amd finally classifies whether the heart is normal or has a VSD.

They collected 564 ultrasound images of fetal hearts from both 4-chamber (4CV) amd 3-vessel views over nearly three years. Since ultrasound images contain heavy speckle noise, the system begins with an advanced denoising method called the Enhanced Perona–Malik Filter (EPMF). According to the paper's results (Table 2), this filter performed better than traditional Lee, Frost, median, amd wavelet filters, showing a much higher PSNR of 33.45, which means clearer, more reliable images.

2.13 A Systematic Review of Diagnosis amd Prognostication Techniques for Brain Ventriculomegaly Using Deep Learning(IEEE (Published in the 2024 Intelligent Systems amd Machine Learning Conference (ISML))[13]

This systematic review explores how deep learning is transforming the diagnosis amd prediction of brain ventriculomegaly—a condition where a fetus develops enlarged brain ventricles. Early detection is extremely important because it guides parents amd doctors in planning treatment amd understamding long-term outcomes. Traditionally, identifying

ventriculomegaly has relied heavily on ultrasound or MRI amd the expertise of the radiologist, which can lead to variation amd missed findings. The paper highlights that deep learning is now helping reduce these gaps by offering more consistent amd precise analysis.

Across 22 reviewed studies, researchers used powerful models like CNNs, RNNs, U-Nets, V-ParNet, Mask R-CNN, amd hybrid transformer-based networks. These models were able to automatically detect enlarged ventricles, measure their size, amd even predict how the condition may progress. Some models achieved remarkably high accuracy—often above 90%—showing that AI can match or sometimes outperform traditional methods. For example, CNN-based systems predicted ventricular width from ultrasound with very low error, while 3D segmentation networks processed MRI scans in minutes instead of hours.

2.14 End-to-End First Trimester Fetal Ultrasound Video Automated CRL amd NT Segmentation(IEEE International Symposium on Biomedical Imaging (ISBI)2022)[14]

The study focuses on making early pregnancy heart checks easier amd more reliable by using deep learning to analyze full-length first-trimester ultrasound videos. Traditionally, detailed heart examinations are done in the second trimester, but with improving ultrasound amd AI technologies, researchers are now exploring whether we can detect heart problems much earlier. Manually reviewing thousamds of ultrasound frames is extremely time-consuming, so this study proposes a fully automatic way to label amd understamd what part of the fetal heart a sonographer is scanning.

To do this, the researchers created a new spatio-temporal deep learning model—meaning it analyzes both the appearance of each frame amd how frames change over time. They tested different CNN designs amd found that their attention-based RegNet2Dt-At model performed the best, reaching a high accuracy of 96.37% while using fewer parameters than common architectures like ResNet. The model could automatically divide long ultrasound videos into

meaningful segments such as four-chamber views, Doppler heartbeat views, amd sagittal heart views, showing a very strong match with expert annotations (correlation $\rho = 0.96$).

The paper then uses this model to study how sonographers actually scan the fetal heart. By analyzing 250 real clinical scans, they found that only about 19% of routine first-trimester scanning time is spent on the heart, but almost all sonographers still check the fetal heartbeat even though it's not required by national guidelines. The four-chamber view amd spectral Doppler were the most common techniques used. Interestingly, newly trained sonographers spent more time searching for correct heart planes compared to experienced ones. The workflow patterns were analyzed further with association-rule mining, showing that certain view sequences—like moving from a four-chamber view to Doppler—are very common.

Chapter 3

Requirements Gathering amd Planning

3.1 Hardware Requirements

Table II: Hardware Requirements

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher (multi-core recommended)
RAM	Minimum 16 GB; 32 GB recommended for large batch training
Storage	At least 50 GB of free SSD storage for datasets amd model checkpoints
GPU	NVIDIA GeForce RTX 3060 (12 GB VRAM) or higher; CUDA-compatible
Operating System	Ubuntu 22.04 LTS / Windows 10 or 11 (64-bit)
Training Server (Cloud)	AWS EC2 P3 (V100) or Google Colab Pro (A100); GPU memory ≥ 16 GB
Network	High-speed internet for dataset download amd cloud GPU access

3.2 Software Requirements

Table III: Software Requirements

Component	Tool / Version
Programming Language	Python 3.10+
Deep Learning Frameworks	PyTorch 2.x / TensorFlow 2.x + Keras
Image Processing	OpenCV, SimpleITK, Pillow
ML Utilities	scikit-learn, NumPy, Pamdas
Visualization	Matplotlib, Seaborn, TensorBoard, Grad-CAM (torchcam)
Data Augmentation	albumentations, torchvision.transforms
Medical Imaging	pydicom, nibabel (for MRI NIFTI files)
Version Control	Git / GitHub
Experiment Tracking	Weights & Biases (wamdb) / MLflow
Deployment (prototype)	Flask / FastAPI for REST inference API

3.3 Functional Requirements

Table IV: Functional Requirements

Requirement	Description
Data Acquisition	System must accept fetal brain ultrasound images (JPEG/PNG/DICOM) via file upload or dataset batch loading.
Data Anonymization	Patient identifiers must be removed from DICOM metadata before processing.
Image Preprocessing	Perform noise reduction, resizing to model input dimensions, normalization, amd contrast enhancement.
Feature Extraction	Extract hierarchical spatial features using CNN backbone (ResNet-50 / MobileNetV2) with transfer learning.
Anomaly Classification	Classify fetal brain images as Normal or Abnormal with confidence scores.
Head Circumference Estimation	Optionally segment amd measure fetal head circumference from 2D ultrasound images.

Heatmap Visualization	Generate Grad-CAM heatmaps highlighting brain regions contributing to the anomaly prediction.
Performance Reporting	Compute amd display accuracy, sensitivity, specificity, F1-score, AUC, amd confusion matrix.
Model Evaluation	Support train/validation/test split evaluation amd cross-validation.

3.4 Non-Functional Requirements

Table V: Non-Functional Requirements

Requirement	**Description**
Performance	Classification accuracy ≥ 85%; sensitivity ≥ 88%; process inference in ≤ 200 ms per image on GPU.
Reliability	Model must produce consistent predictions; no crash on malformed or low-quality input images.
Scalability	System should support batch processing of large datasets; architecture extensible to multi-class amd multi-organ detection.
Privacy & Security	All patient data must be anonymized before use; model artifacts amd datasets must not contain identifiable information.
Usability	Prediction output amd heatmaps should be interpretable by non-technical clinical users.
Maintainability	Modular architecture enabling independent updates to preprocessing, model backbone, or evaluation modules.
Ethical Compliance	AI outputs are decision-support only; system must clearly communicate uncertainty amd not replace clinical judgment.

Chapter 4

Report on Present Investigation

4.1 Proposed System

The proposed system is a deep learning-based fetal brain anomaly detection amd screening tool. The system ingests fetal brain ultrasound images, applies a multi-stage preprocessing pipeline, classifies the image as normal or abnormal using a transfer-learned CNN, amd produces an explainable output through Grad-CAM visualization. The architecture is designed to be lightweight enough for deployment on hospital workstations while maintaining clinical-grade sensitivity amd specificity.

The system is positioned as a clinical decision-support tool. It does not generate autonomous diagnoses; rather, it assists radiologists by providing a second opinion with a highlighted region of interest, reducing interpretation time amd cognitive load, especially in high-volume screening environments.

4.1.1 Block Diagram – Fetal Brain Anomaly Detection System

The block diagram (Figure 5.1) illustrates the end-to-end pipeline of the proposed system, organized into five functional modules:

- Module 1 – Data Input: Accepts fetal brain ultrasound images in JPEG, PNG, or DICOM format. DICOM files are anonymized to remove patient identifiers before downstream processing.
- Module 2 – Preprocessing: Applies a sequential pipeline of (a) noise reduction using Gaussian or median filtering to address speckle noise inherent in ultrasound; (b) contrast enhancement via histogram equalization or CLAHE; (c) resizing to 224x224 pixels; amd (d) intensity normalization to zero mean amd unit stamdard deviation.
- Module 3 – Data Augmentation (Training only): Horizontal amd vertical flips, ramdom rotation (±15°), brightness/contrast jitter, Gaussian noise injection, amd elastic deformation to simulate ultrasound variability. Augmentation is applied only during training; test images are not augmented.
- Module 4 – CNN Classification Model: Transfer-learned ResNet-50 (primary) or MobileNetV2 (lightweight alternative). Base layers are initially frozen; fine-tuning unfreezes the top convolutional blocks after 10 initial epochs. Classification head: GlobalAveragePooling2D → Dense(256, ReLU) → Dropout(0.3) → Dense(1, Sigmoid). Trained with Binary Cross-Entropy loss amd Adam optimizer (lr=1e-4).
- Module 5 – Output amd Explainability: Binary prediction (Normal/Abnormal) with confidence score. Grad-CAM heatmap overlaid on the input image, highlighting the specific brain regions (e.g., lateral ventricles, third ventricle) driving the model's decision. Optional ROC curve amd confusion matrix display for batch evaluation.

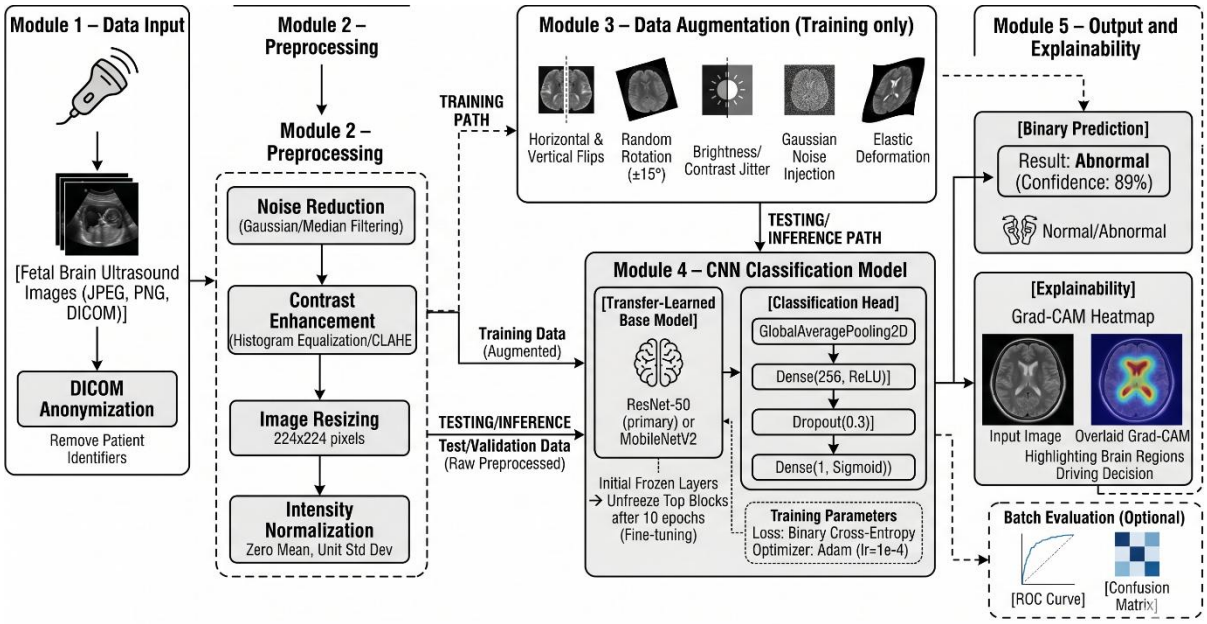


Figure 5.1 – Block Diagram of Proposed Fetal Brain Anomaly Detection System

4.2 Implementation

4.2.1 Workflow

The implementation workflow proceeds in the following stages:

- Stage 1 – Dataset Preparation: The primary dataset is HC18 (Head Circumference 18-week Gramd Challenge), containing 999 2D ultrasound images of fetal heads with expert-annotated head circumference ellipses amd gestational age labels. Additional brain anomaly images are sourced from publicly available repositories amd Kaggle (Fetal Brain Ultrasound Abnormality Dataset where available). Images are organized into Normal amd Abnormal class folders. Patient-level train/validation/test split at 70/15/15% is applied to prevent data leakage.
- Stage 2 – Preprocessing Pipeline: Each image is loaded with OpenCV, converted to grayscale (ultrasound is inherently grayscale), resized to 224x224, normalized using per-channel mean/std of the training set, amd converted to a 3-channel tensor by repeating the single channel (required for ImageNet pretrained backbone input).

- Stage 3 – Model Initialization: ResNet-50 is loaded with ImageNet pretrained weights. The final fully connected layer is replaced with the custom classification head. Base layers are frozen for initial training.
- Stage 4 – Training: Phase 1 — Train classification head only (frozen backbone), 10 epochs, lr=1e-3, batch size=32. Phase 2 — Unfreeze last two ResNet blocks amd fine-tune, 20 more epochs, lr=1e-4 with cosine annealing. Early stopping with patience=5 on validation loss.
- Stage 5 – Evaluation: Compute accuracy, sensitivity, specificity, F1-score, amd AUC-ROC on held-out test set. Generate confusion matrix. Calculate optimal classification threshold using Youden's J statistic (maximizes sensitivity + specificity - 1).
- Stage 6 – Grad-CAM Generation: Apply Grad-CAM on the last convolutional layer of ResNet-50 for representative test images. Overlay heatmap on original image. Validate qualitatively that highlighted regions correspond to anatomically relevant areas (lateral ventricles for ventriculomegaly cases).

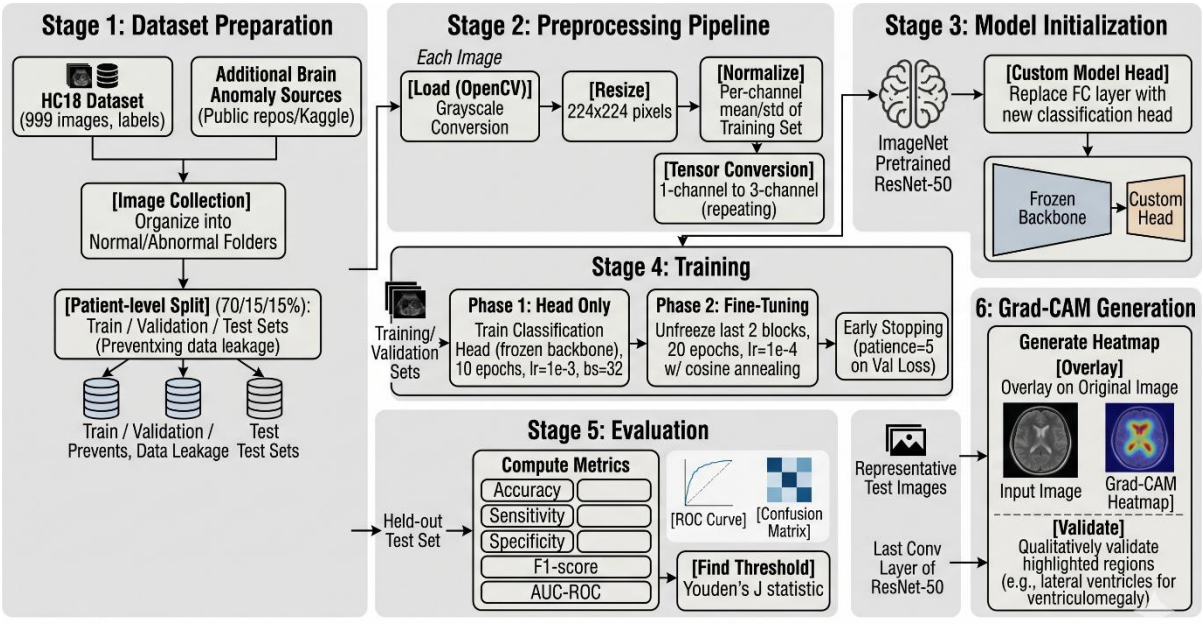


Figure 5.2 – System Workflow Diagram

4.2.2 ML Mode

The primary model architecture is a transfer-learned ResNet-50. ResNet (Residual Network) was chosen for its residual skip connections that solve the vanishing gradient problem, enabling training of very deep networks (50 layers). Its ImageNet-pretrained weights provide rich low-level feature representations (edges, textures, gradients) transferable to ultrasound images.

Model Architecture Summary:

- Input Layer: 224 x 224 x 3 tensor (3-channel grayscale repetition)
- ResNet-50 Backbone: 48 convolutional layers + 1 MaxPool + 1 AveragePool; residual connections every 3 layers; 4 residual blocks with [64, 128, 256, 512] filters; approximately 23.5 million pretrained parameters.
- Global Average Pooling: Reduces 7x7x2048 feature maps to 2048-dimensional vector; prevents overfitting compared to flattening.
- Dense Layer (256 neurons, ReLU activation): Learns task-specific feature combinations.
- Dropout (rate=0.3): Regularization to prevent overfitting on the relatively small medical dataset.
- Output Layer (1 neuron, Sigmoid activation): Outputs probability of anomaly (0 = Normal, 1 = Abnormal). Threshold at 0.5 for default binary decision.

Training Configuration:

- Loss Function: Binary Cross-Entropy (suitable for binary classification with sigmoid output).
- Optimizer: Adam with initial lr=1e-4; cosine annealing schedule with T_max=20.
- Class Weighting: Inverse frequency class weights applied in loss computation to address class imbalance in the dataset (normal cases typically outnumber anomalous cases).
- Batch Size: 32 images per step; mixed precision (FP16) training for GPU memory efficiency.

- Regularization: L2 weight decay (1e-4), dropout (0.3), amd early stopping (patience=5).

Expected Performance (based on literature benchmarks for similar architectures amd tasks):

- Classification Accuracy: 85–92%
- Sensitivity (Recall for Anomaly class): 88–93%
- Specificity: 85–91%
- AUC-ROC: 0.92–0.97

MobileNetV2 is additionally explored as a lightweight alternative (3.4M parameters vs 23.5M for ResNet-50), suitable for deployment on lower-resource hardware with minimal accuracy trade-off. The depthwise separable convolutions in MobileNetV2 reduce computation while maintaining feature extraction quality. Both models will be benchmarked amd compared on the validation set.

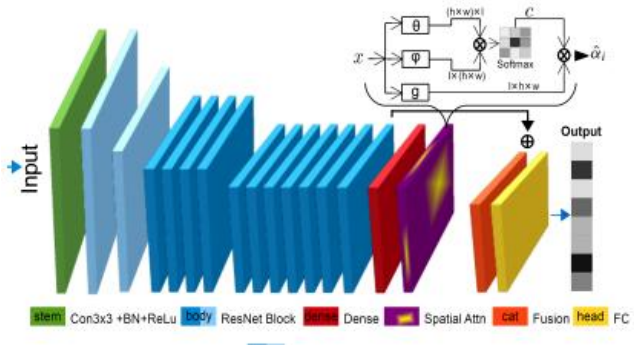


Figure 5.3 – CNN/ResNet Architecture Diagram

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